

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct:

I JITESH AP (192511163) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

CO₂ - AT₃

Debugging and optimisation Task

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CSA0707 - Computer
Networks

①

Given :

* IP: 192.168.10.65

* Subnet Mask : 255.255.255.192 (126)

* Gateway : 192.168.10.1

* Subnet of configured IP

192.168.10.65 belongs to 192.168.10.64/26

* network details

network address : 192.168.10.0

Broadcast address : 192.168.10.63

Valid Host range : 192.168.10.1 - 192.168.10.62

* verify

Ip address : Invalid

Gateway : Invalid

Issue:

no prefix mismatch; communication failure is likely due to another configuration issue.

optimized address:

Device A: 2001:db8:0:0::10/64

Device B: 2001:db8:0:0::20/64

* Available Host Address

$2^{64} = 18,446,744,073,709,551,616$ addresses

(3) Given:

Network: 192.168.1.0/24

Subnet mask: /26, /27, /28

1. Identify Subnet Details

Subnet	Network address	Broadcast Address	Host range	Usable Hosts
/26	192.168.1.0	192.168.1.63	192.168.1.1 - 192.168.1.62	62
/27	192.168.1.64	192.168.1.95	192.168.1.65 - 192.168.1.94	30
/28	192.168.1.96	192.168.1.111	192.168.1.97 - 192.168.1.110	14

2. calculate unused IP addresses:

$2^{1/26}$: 62 usable host

$2^{1/27}$: 30 usable host

$2^{1/28}$: 14 usable host.

3. Identify inefficient subnet allocation:

Large department should use $1/26$

Medium department should use $1/27$.

Small department use $1/28$.

4. Debug The addressing plan:

Check for overlapping subnet

Ensure Each department has a unique subnet

Verify network and broadcast addresses are not assigned to host.

5. Optimize using VLSM

Assign subnets according to department size:

large $\rightarrow 1/26$

medium $\rightarrow 1/27$

Small $\rightarrow 1/28$

2. reduce IP wastage & Improves network utilization.

Expand IPv6 Address

A : 2001 : db8 : 0000 : 0000 : 0000 : ff00 : 00 42 : 8 329

B : 2001 : db8 : 0000 : 0000 : 0001 : 0000 : 0000 : 0001

* Network prefix

Both : 2001 : db8 : 0 : 0 : : /64

* Subnet check :

both devices are in same subnet

* Error

Interface or routing configuration

* optimized address :

Device A : 2001 : db8 : 00 : : 10/64

Device B : 2001 : db8 : 0 : 0 : : 20/64

* Available hosts :

2^{64} host addresses

(5) Given Routing Table:

$192.168.1.0/24 \rightarrow 10.0.0.2$

$192.168.2.0/24 \rightarrow 10.0.0.3$

Default Route $\rightarrow 10.0.0.1$

1. largest prefix match:

Destination $192.168.2.100$ matches $192.168.2.0/24$

2. next hop router

$10.0.0.3$

3. verify destination subnet

$192.168.2.100$ belongs to $192.168.2.0/24$

4. Routing Error:

If packet fails check the route to $10.0.0.3$ or
verify the routing table configuration

5. Optimization :

Ensure route $192.168.2.0/24 \rightarrow 10.0.0.3$ is correct
and remove any incorrect or duplicate routes.

6. Default route

If no destination matches, packets are forwarded
to $10.0.0.1$